

# SCISSOR: SCORE-CONDITIONED INSTRUMENT SOURCE SEPARATION FOR ORCHESTRAL RECORDINGS

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## ABSTRACT

Orchestral separation recovers instrument sections from mixtures in which shared pitches, harmonics, and timbres obscure source identity. An aligned score provides instrument labels, note pitches, and activity times. A score-informed approach appends piano rolls to audio features before mask prediction. We introduce *SCISSOR* (Score-Conditioned Instrument Source Separation for Orchestral Recordings), which uses the score to form a frame-wise query for each source. Each query matches a shared audio representation, and a softmax over instrument and background slots jointly assigns overlapping time–frequency evidence. The queries retain instrument identity even when notes are missing from the score. After training on SynthSOD and a small set of URMP and PHENICX-Anechoic recordings, *SCISSOR* achieves the highest average SDR on held-out real recordings. With SynthSOD-only training, it leads on SynthSOD and zero-shot PHENICX-Anechoic, and improves on its audio-only control on zero-shot URMP. *SCISSOR* also degrades less under score corruption than the evaluated score-based baselines.

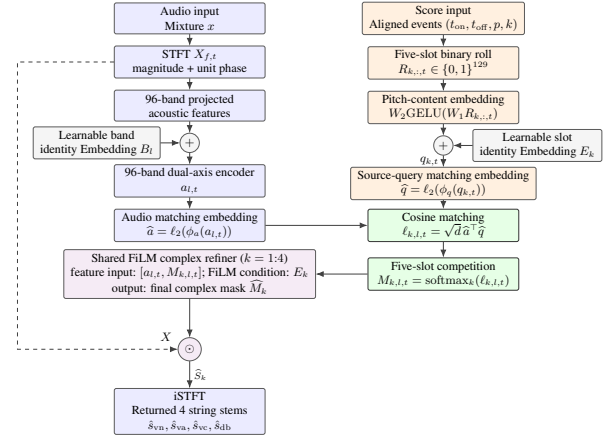
**Index Terms**— Music source separation, Score-informed separation, Orchestral music

## 1. INTRODUCTION

Music source separation (MSS) recovers individual sources from a music recording [1]. In orchestral MSS, these sources are instrument sections, rather than the vocal, bass, drum, and accompaniment stems often studied in popular music [2]. Sections may play the same pitch and produce similar timbres and overlapping harmonics. As a result, the mixture alone may not reveal which section produced a given sound [3].

Before deep learning, audio-only MSS commonly used non-negative matrix factorization (NMF) to decompose mixture spectrograms, with constraints on temporal continuity and sparsity [4]. Neural separators learn source estimates from data. Spectrogram-based models predict masks with convolutional networks [5], recurrent networks such as X-UMX [6], or subband models such as BSRNN and BS-RoFormer [7, 8]. HTDemucs combines spectral and waveform representations [9]. Despite these advances, audio-only models must infer instrument activity and identity from the mixture itself, which is difficult when orchestral sources overlap in pitch and timbre.

An aligned score specifies each instrument’s note pitches and onset and offset times. Earlier score-informed methods used these cues to constrain NMF [10, 11] or to guide neural training with score-filtered spectra and weak labels [12, 13]. More recently, Tunturi et al. [14] encoded instrument piano rolls and concatenated them with the mixture spectrogram in an X-UMX separator. They



**Fig. 1.** SCISSOR overview: score-derived source queries compete for shared audio evidence, and a shared refiner reconstructs four string stems.

also tested a score-only variant that predicts masks without audio features. Score concatenation helped on small ensembles, and the score-only variant transferred well to some real recordings. In denser orchestral mixtures, however, the gains from concatenation were less consistent, and the score-informed model struggled to transfer from synthetic to real audio. The effect of score misalignment was not examined [14].

We propose *SCISSOR*, a score-conditioned source-query method for orchestral separation. A shared encoder represents the mixture across time and frequency. At each frame, *SCISSOR* combines active score pitches with a learned instrument identity to form a query for each source. Even if the score omits a note, the identity remains in the query and keeps that source available. Separate projections map audio features and score queries into a shared space, where their normalized embeddings are matched at each band–frame location. A softmax over four instrument slots and a background slot then assigns mixture evidence jointly: a stronger match for one source leaves less evidence for the others. A shared complex refiner uses these soft assignments and the audio features to reconstruct the instrument stems. Joint query competition lets acoustic evidence temper inaccurate score cues.

**Our contributions are as follows:**

- **A new score-informed separation method.** *SCISSOR* uses frame-wise instrument queries to match score cues with encoded audio and allocate mixture evidence jointly across sources.
- **Improved separation quality.** *SCISSOR* achieves the highest average SDR among the evaluated systems on held-out

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real recordings and SynthSOD.

- **Effective use of limited real data and robust score guidance.** SCISSOR performs well with a small real-recording training set and loses less SDR under score corruption than the evaluated score-based baselines.

We train SCISSOR on aligned scores and audio from synthetic SynthSOD and real URMP and PHENICX-Anechoic recordings [2, 15, 16]. We evaluate separation on held-out synthetic and real recordings, zero-shot transfer to real audio, and robustness to score corruption.

Audio examples are available at <https://scissorsep.github.io/>. All project code and model checkpoints will be made publicly available upon acceptance.

## 2. METHOD

SCISSOR extracts four string sections from a full-orchestra mixture, as shown in Fig. 1. Given a mono mixture  $x$  and an aligned symbolic score  $\mathcal{S}$ , the model estimates violin, viola, cello, and double-bass signals ( $K = 4$ ). The audio pathway computes the log magnitude and normalized phase of the mixture spectrum  $X$ , projects these features onto  $L = 96$  overlapping quasi-log-frequency bands, and adds a learned band identity. Dual-axis attention then produces an audio representation  $a_{l,t}$  for band  $l$  and frame  $t$ . The score pathway maps note events to four target slots and one background slot, rasterizes pitch activity on the audio frame grid, and combines each activity vector with a learned slot identity. Separate audio and score projections yield scaled cosine compatibilities; a softmax over the five slots jointly allocates each band-frame region. A shared identity-conditioned complex refiner uses the four target allocations and audio features to estimate complex masks  $\widehat{M}_k$  and sources  $\widehat{S}_k = \widehat{M}_k \odot X$ . The background allocation absorbs non-target energy internally. The following components specify the query construction, matching, and source assignment.

### Frame-wise source queries.

Each score event specifies an onset, offset, pitch, and MIDI program. Events for the four requested string sections enter their respective target slots; other programs enter the background slot. We dilate onsets by 30 ms and offsets by 80 ms, then rasterize the events on the audio frame grid as  $R \in \{0, 1\}^{K' \times 129 \times T}$ , where  $K' = K + 1$ . A source query combines the active pitches of slot  $k$  with its learned identity  $E_k$ :

$$q_{k,t} = E_k + W_2 \text{GELU}(W_1 R_{k,:t}). \quad (1)$$

The vector  $R_{k,:t} \in \{0, 1\}^{129}$  records the pitches of source  $k$  at frame  $t$ ;  $W_1$  and  $W_2$  embed that activity, and  $E_k \in \mathbb{R}^d$  identifies the source. Thus  $q_{k,t}$  represents both the expected instrument and its active pitches. Because the two linear layers have no bias, an empty pitch vector yields  $q_{k,t} = E_k$ . The instrument therefore remains a candidate when the score omits a note or misaligns its timing.

### Cross-modal matching embeddings.

SCISSOR projects audio features and score queries separately into a shared matching space:

$$\begin{aligned} \phi_r(z) &= V_r \text{GELU}(U_r \text{LN}(z) + b_r), \quad r \in \{a, q\}, \\ \widehat{a}_{l,t} &= \ell_2(\phi_a(a_{l,t})), \quad \widehat{q}_{k,t} = \ell_2(\phi_q(q_{k,t})). \end{aligned} \quad (2)$$

The index  $r \in \{a, q\}$  selects the audio or query projection, and  $\ell_2(v) = v/\|v\|_2$ . Separate projections accommodate the input statistics of the two modalities. Unit normalization makes  $\widehat{a}_{l,t}^\top \widehat{q}_{k,t}$  a

cosine compatibility between the shared audio representation and a source query.

### Score-conditioned instrument source separation.

The  $K' = K + 1$  queries comprise four requested string sections and a background slot containing non-target score programs. For each audio band-frame representation, SCISSOR computes query compatibilities and normalizes them over the source axis:

$$M_{k,l,t} = \text{softmax}_k \left( \sqrt{d} \widehat{a}_{l,t}^\top \widehat{q}_{k,t} \right). \quad (3)$$

The fixed factor  $\sqrt{d}$  scales cosine compatibility, and  $M_{k,l,t}$  denotes the allocation to source  $k$  at band  $l$  and frame  $t$ . The softmax enforces  $\sum_{k=1}^{K'} M_{k,l,t} = 1$ : a larger allocation to one source reduces the allocation available to its competitors. The background slot absorbs non-target energy without producing an output stem. Because the allocations remain soft, acoustic evidence can counter an absent or incorrect score event.

### Training and implementation details.

We train SCISSOR end to end with

$$\mathcal{L} = \mathcal{L}_{\text{wav}} + \mathcal{L}_{\text{cSTFT}} + 0.25\mathcal{L}_{\text{mask}} + 0.15\mathcal{L}_{\text{comp}} + 0.15\mathcal{L}_{\text{SI}}, \quad (4)$$

The five terms measure waveform  $\ell_1$ , complex-STFT  $\ell_1$ , band-ratio-mask error, competition-region error, and scale-invariant waveform error, respectively. A small loss weight on silent targets discourages leakage. For each requested source, a shared FiLM-conditioned refiner takes  $[a_{l,t}, M_{k,l,t}]$ , predicts real and imaginary band-domain corrections, and converts them to a bounded complex mask. The model uses  $d = 448$ , a 256-dimensional refiner state, and nine dual-axis audio blocks. Unless specified otherwise, experiments use 44.1-kHz audio, a 4096-point STFT with hop size 1024, 96 bands, and the score augmentations in Section 3.

## 3. EXPERIMENT

### SynthSOD.

SynthSOD [2, 14] contains synthesized ensembles and orchestras with isolated stems spanning 18 instrument tracks: strings, woodwinds, brass, harp, and percussion. Our 484-piece subset has two to 14 active tracks per piece and aligned score events specifying each note’s onset, offset, pitch, and instrument. We use 351 pieces for training, 42 for validation, and 91 for testing. The input is the mono full mixture from the Tree microphone. Violin, viola, cello, and double bass are the targets; all other stems remain in the mixture. Among the test pieces, 59 contain at least one target instrument.

### URMP.

URMP [15] contains 44 real chamber-music recordings with two to five parts per piece, covering 13 instrument types across strings, woodwinds, and brass. Its score and note annotations are aligned with the recorded parts; we use note onset, offset, pitch, and instrument labels. We select the 22 recordings with string targets and split them by musical work into 13 training, four validation, and five test pieces. All four requested string instruments are evaluated when present.

### PHENICX-Anechoic.

PHENICX-Anechoic [16] contains four real orchestral excerpts with 10 to 39 separately recorded parts per piece. They cover up to ten instrument types across strings, woodwinds, and brass. The aligned score provides note onset, offset, pitch, and instrument labels, alongside isolated audio references. We use two pieces for training, one for validation, and one for testing. Across URMP and PHENICX-Anechoic, the six test pieces come from three works absent from training and validation.

**Table 1.** Museval SDR (dB) on held-out real recordings after SynthSOD and real-recording training. Values are estimates [95% CI]; the one-piece PHENICX test has no across-recording CI.

| System                                       | Vn.         | Va.         | Vc.          | Db.         | Avg. [95% CI]            |
|----------------------------------------------|-------------|-------------|--------------|-------------|--------------------------|
| <b>URMP (<math>n = 5</math>)</b>             |             |             |              |             |                          |
| HTDemucs                                     | 0.23        | 0.57        | 5.07         | 6.13        | 3.00[ 0.25, 5.09]        |
| BS-RoFormer                                  | 2.46        | 0.86        | 3.31         | 5.32        | 2.99[ 1.00, 3.51]        |
| Tunturi et al., baseline                     | 1.91        | 1.43        | 5.52         | 5.89        | 3.69[ 1.63, 4.48]        |
| Tunturi et al., score-informed               | 2.27        | 1.76        | 5.86         | 5.86        | 3.94[ 1.66, 5.38]        |
| Tunturi et al., score-only                   | 4.90        | 3.50        | 6.69         | 7.40        | 5.62[ 4.09, 7.17]        |
| SCISSOR (audio only)                         | 2.21        | 2.30        | 6.64         | 7.65        | 4.70[ 1.93, 6.60]        |
| <b>SCISSOR (ours)</b>                        | <b>6.10</b> | <b>4.49</b> | <b>7.81</b>  | <b>8.81</b> | <b>6.80[ 4.69, 8.98]</b> |
| <b>PHENICX-Anechoic (<math>n = 1</math>)</b> |             |             |              |             |                          |
| HTDemucs                                     | -1.87       | -0.23       | -10.83       | -3.56       | -4.12                    |
| BS-RoFormer                                  | -2.42       | 0.00        | -1.39        | 5.10        | 0.32                     |
| Tunturi et al., baseline                     | -0.38       | -2.23       | -2.95        | 3.40        | -0.54                    |
| Tunturi et al., score-informed               | -0.73       | -2.28       | -2.95        | 4.83        | -0.29                    |
| Tunturi et al., score-only                   | 0.07        | -4.33       | -4.08        | 1.84        | -1.62                    |
| SCISSOR (audio only)                         | 1.00        | -0.09       | <b>-0.01</b> | 5.67        | 1.64                     |
| <b>SCISSOR (ours)</b>                        | <b>1.28</b> | <b>0.10</b> | -0.02        | <b>6.00</b> | <b>1.84</b>              |

#### Implementation details.

SCISSOR processes 44.1-kHz mono audio with a 4096-point STFT and a 1024-sample hop. Log-magnitude and unit-phase features are projected onto 96 frequency bands; audio and score queries use 448-dimensional embeddings. SCISSOR has 28.04 million parameters. We also train an audio-only SCISSOR with the same architecture and parameter count. It receives an empty score during training, validation, and inference, leaving only the learned instrument identities in its source queries; its data, loss, optimizer, and training budget match those of SCISSOR. We train on four NVIDIA RTX 4080 GPUs. SynthSOD training runs from random initialization for 20,000 optimizer steps on 4-s crops with global batch size 12. We use AdamW with learning rate  $1.5 \times 10^{-4}$ , 1,500 warm-up steps, cosine decay, and an exponential moving average of 0.999. Real recording training starts from the SynthSOD checkpoint and runs for 2,000 steps with effective batch size 12. During training, we randomly withhold the score, jitter note onsets, and drop pitch cells.

#### Evaluation metrics.

We measure separation quality with signal-to-distortion ratio (SDR, in dB); higher values indicate better separation. Tables 1 and 2 use museval BSSEval v4 SDR [17], computed in one-second windows. For each instrument, we take the median across windows within each recording and then the median across recordings. Avg. is the unweighted mean of the four instrument scores; silent references are excluded. Table 3 instead uses whole-piece energy-ratio SDR, averaged over active instruments within each piece.

For aggregates over at least two recordings or pieces, we report 95% confidence intervals (CIs) from the 2.5th and 97.5th percentiles of 10,000 bootstrap replicates (seed 20260924). We resample whole recordings or pieces with replacement and recompute the target-median macro-average for Tables 1 and 2, or the mean of piece-level target averages for Table 3. One-second windows are not treated as independent samples. Results are shown as *estimate* [95% CI]; the single held-out PHENICX-Anechoic recording has no across-recording CI.

#### Baseline models and comparison setup.

We compare SCISSOR with HTDemucs [9] (41.98M parameters), BS-RoFormer [8] (58.07M), and the audio-only (28.32M), score-informed (29.77M), and score-only (25.50M) X-UMX models of

Tunturi et al. [14], as well as the matched audio-only SCISSOR (28.04M). All systems use the aligned SynthSOD training and validation sets. For Table 1, they then train on the same aligned URMP and PHENICX-Anechoic recordings. Audio-only models receive audio without score cues. HTDemucs and BS-RoFormer use the same SynthSOD 351/42 split and training budget as SCISSOR, while the X-UMX models start from released SynthSOD checkpoints. We use common mixtures and scoring code and evaluate only the violin, viola, cello, and double-bass outputs.

#### Real-recording separation test setup (Table 1).

Every model is first trained on SynthSOD, then trained on 15 aligned real recordings (13 URMP and two PHENICX-Anechoic), with five others (four URMP and one PHENICX-Anechoic) used for model selection. We then evaluate separation on five held-out URMP recordings and one held-out PHENICX-Anechoic recording, reporting the two corpora separately.

#### Synthetic and zero-shot separation test setup (Table 2).

Using only SynthSOD-trained checkpoints, we compare the same systems on the 59 test pieces with string targets. We then test transfer to all 22 URMP and four PHENICX-Anechoic recordings without real-recording training. The unseparated mixture provides a performance floor.

#### Score-corruption test setup (Table 3).

We compare SCISSOR with Tunturi et al.’s score-informed and score-only models on 22 SynthSOD test pieces containing all four strings. Across 33 conditions, we perturb note timing, pitch, presence, instrument identity, and score structure, including combined errors. The displayed severe conditions include 400-ms onset jitter, 50% pitch substitution, 75% note deletion, and complete instrument relabelling. For each condition, the checkpoint, mixture, references, and inference procedure remain fixed; only the score changes. Clean scores, withheld scores, and matched audio-only models serve as controls.

## 4. RESULTS

#### Real-recording separation test results.

Table 1 compares systems trained on SynthSOD and real record-

**Table 2.** Avg. museval SDR (dB) after SynthSOD-only training; estimates with 95% recording-bootstrap CIs.

| System                         | Score | SynthSOD ( $n = 59$ )                     | PHENICX ( $n = 4$ )                       | URMP ( $n = 22$ )                         |
|--------------------------------|-------|-------------------------------------------|-------------------------------------------|-------------------------------------------|
| Mixture (no separation)        |       | -7.45 [ -8.69 , -6.36 ]                   | -14.11 [ -17.80 , -8.26 ]                 | -5.23 [ -5.72 , -3.38 ]                   |
| HTDemucs [9]                   |       | 4.17 [ 2.43 , 5.20 ]                      | -1.83 [ -5.74 , 1.20 ]                    | 1.93 [ 0.75 , 3.31 ]                      |
| BS-RoFormer [8]                |       | 3.39 [ 2.30 , 4.18 ]                      | -2.72 [ -5.33 , 0.16 ]                    | 1.54 [ 0.84 , 2.63 ]                      |
| Tunturi et al., baseline [14]  |       | 5.64 [ 4.23 , 6.27 ]                      | -0.15 [ -1.51 , 2.35 ]                    | 1.88 [ 0.99 , 2.69 ]                      |
| Tunturi et al., score-informed | ✓     | 5.96 [ 4.59 , 6.53 ]                      | 0.55 [ -0.69 , 2.35 ]                     | 3.35 [ 2.17 , 3.94 ]                      |
| Tunturi et al., score-only     | ✓     | 4.35 [ 3.40 , 5.01 ]                      | 0.83 [ -2.43 , 3.39 ]                     | <b>4.80</b> [ <b>4.13</b> , <b>5.96</b> ] |
| SCISSOR (audio only)           |       | 5.97 [ 4.74 , 6.56 ]                      | 1.43 [ 0.18 , 2.50 ]                      | 1.88 [ 1.21 , 2.47 ]                      |
| <b>SCISSOR (ours)</b>          | ✓     | <b>6.31</b> [ <b>5.15</b> , <b>6.90</b> ] | <b>1.85</b> [ <b>1.16</b> , <b>2.78</b> ] | 2.52 [ 1.86 , 3.34 ]                      |

**Table 3.** Whole-piece SDR (dB) under score corruption on 22 SynthSOD test pieces; estimates with 95% piece-bootstrap CIs.

| Score error                       | Tunturi score-informed [14] | Tunturi score-only   | SCISSOR           |
|-----------------------------------|-----------------------------|----------------------|-------------------|
| Clean score                       | 4.81 [4.50, 5.11]           | 2.92 [2.49, 3.33]    | 5.45 [5.13, 5.76] |
| Matched audio-only control        | 4.58 [4.29, 4.89]           | 4.58 [4.29, 4.89]    | 4.98 [4.67, 5.29] |
| Onset jitter ( $\sigma = 400$ ms) | 4.38 [4.10, 4.67]           | 2.15 [1.79, 2.50]    | 5.23 [4.91, 5.53] |
| Global offset (1 s)               | 3.92 [3.64, 4.21]           | 1.78 [1.45, 2.10]    | 4.97 [4.67, 5.27] |
| Tempo scaling (10%)               | 3.35 [3.10, 3.61]           | 0.99 [0.75, 1.22]    | 4.72 [4.42, 5.01] |
| Pitch substitution (50%)          | 4.34 [4.04, 4.63]           | 2.10 [1.70, 2.46]    | 5.29 [4.98, 5.59] |
| Note deletion (75%)               | 3.40 [3.14, 3.66]           | 0.73 [0.53, 0.92]    | 5.04 [4.72, 5.35] |
| Missing parts (two)               | 3.77 [3.44, 4.11]           | 1.11 [0.62, 1.60]    | 5.19 [4.88, 5.49] |
| Instrument relabelling (100%)     | 2.70 [2.42, 2.98]           | -0.58 [-0.97, -0.26] | 3.92 [3.65, 4.19] |
| Part swaps (both pairs)           | 2.14 [1.73, 2.49]           | -0.60 [-1.05, -0.21] | 3.45 [3.12, 3.75] |
| Structural cut (20%)              | 3.75 [3.46, 4.06]           | 1.35 [1.06, 1.68]    | 5.00 [4.66, 5.32] |
| Combined, severe                  | 4.05 [3.78, 4.32]           | 1.71 [1.40, 2.01]    | 5.04 [4.73, 5.33] |
| Score withheld                    | 2.87 [2.60, 3.15]           | 0.09 [0.06, 0.12]    | 4.96 [4.64, 5.26] |

ings. On five held-out URMP pieces, SCISSOR reaches 6.80 dB Avg. SDR [4.69, 8.98], a 1.18-dB gain over the strongest external baseline, Tunturi et al.’s score-only model at 5.62 dB [4.09, 7.17]. Relative to the best external baseline for each instrument, the gains are 1.20, 0.99, 1.12, and 1.41 dB for violin, viola, cello, and double bass, respectively. On the held-out PHENICX-Anechoic recording, SCISSOR improves Avg. SDR by 1.52 dB over the best external baseline (1.84 versus 0.32 dB); the corresponding instrument gains are 1.21, 0.10, 1.37, and 0.90 dB. Its cello result is 0.01 dB below its own audio-only control. The URMP confidence intervals overlap, so these gains do not establish statistical significance. The single PHENICX test piece has no across-recording confidence interval.

#### Synthetic and zero-shot separation test results.

Table 2 uses checkpoints trained only on SynthSOD. On the 59-piece synthetic test, SCISSOR has the highest Avg. SDR at 6.31 dB [5.15, 6.90], compared with 5.96 dB [4.59, 6.53] for the strongest external baseline. It also has the highest point estimate on four zero-shot PHENICX recordings, 1.85 dB [1.16, 2.78], although its interval overlaps that of its audio-only control at 1.43 dB [0.18, 2.50]. On 22 zero-shot URMP recordings, SCISSOR improves on its control (2.52 versus 1.88 dB), but Tunturi et al.’s score-only model remains higher at 4.80 dB [4.13, 5.96]. The SynthSOD intervals also overlap, whereas the URMP intervals for SCISSOR [1.86, 3.34] and the score-only model do not. Thus, the synthetic-data advantage does not extend to every recorded ensemble.

#### Score-corruption test results.

Table 3 shows representative errors from 33 score corruptions. With 75% of notes deleted, SCISSOR loses 0.41 dB relative to its clean score, compared with losses of 1.41 dB for Tunturi et al.’s score-

informed model and 2.19 dB for its score-only model. The resulting SDRs are 5.04 [4.72, 5.35], 3.40 [3.14, 3.66], and 0.73 dB [0.53, 0.92], with non-overlapping 95% CIs. Complete instrument relabelling causes larger drops of 1.53, 2.11, and 3.50 dB, respectively; SCISSOR still has the highest SDR at 3.92 dB [3.65, 4.19], compared with 2.70 dB [2.42, 2.98] and -0.58 dB [-0.97, -0.26]. Across all 33 conditions, SCISSOR falls below its audio-only control in 8, versus 23 for score-informed and 33 for score-only separation. Instrument-identity errors remain challenging: relabelling and part swaps reduce SCISSOR below its 4.98-dB audio-only control.

## 5. CONCLUSION

SCISSOR forms frame-wise source queries from score pitches and instrument identities, matches them to shared audio features, and jointly assigns mixture energy to instrument and background slots. Identity-only queries keep instruments available when notes are omitted. With a small real-recording training set, SCISSOR led the evaluated systems on held-out URMP and PHENICX-Anechoic recordings. Trained only on SynthSOD, it led on SynthSOD and zero-shot PHENICX-Anechoic and surpassed its audio-only control on zero-shot URMP; Tunturi et al.’s score-only model led on the latter. SCISSOR degraded less under representative score errors, although instrument-identity errors remain challenging.

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## COMPLIANCE WITH ETHICAL STANDARDS

This study used only publicly available music datasets and involved no new human or animal subject experiments; no ethics approval was required.

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